

Module 11: Week 13: Required Assignment

1. Understanding NoSQL Databases (8 Marks)

a) Define NoSQL databases and describe their key features

NoSQL databases are non-relational databases or sometimes “Not-only SQL” databases if it supports SQL queries. Its key features are flexible schema and horizontal scalability.

b) Compare NoSQL vs SQL databases by creating a table with at least 4 parameters

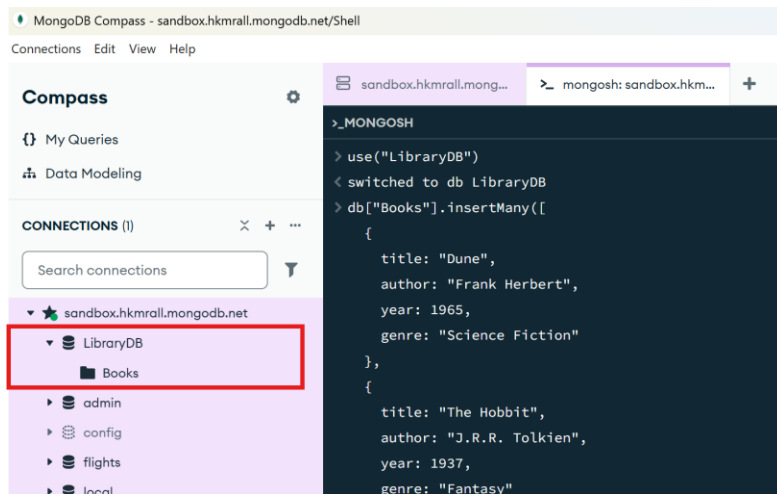
Parameter	SQL	NoSQL
Schema	Fixed, predefined schema; all rows in a table share the same structure, type of the column is declared at table creation	Dynamic/flexible schema; structure can vary per record, each key can have different types
Scalability	Vertical	Vertical & horizontal
Data Storage	Tables with rows and columns; relational	Documents, key-value pairs, non-relational
Data Retrieval	Even a single-row fetch needs to go through relational overhead	Optimised for known access patterns with no JOINS
Consistency	ACID (Atomicity, Consistency, Isolation, Durability)	BASE (Basically Available, Soft state, Eventual consistency)

c) Mention 2 real-world use cases where NoSQL databases are more suitable than SQL databases

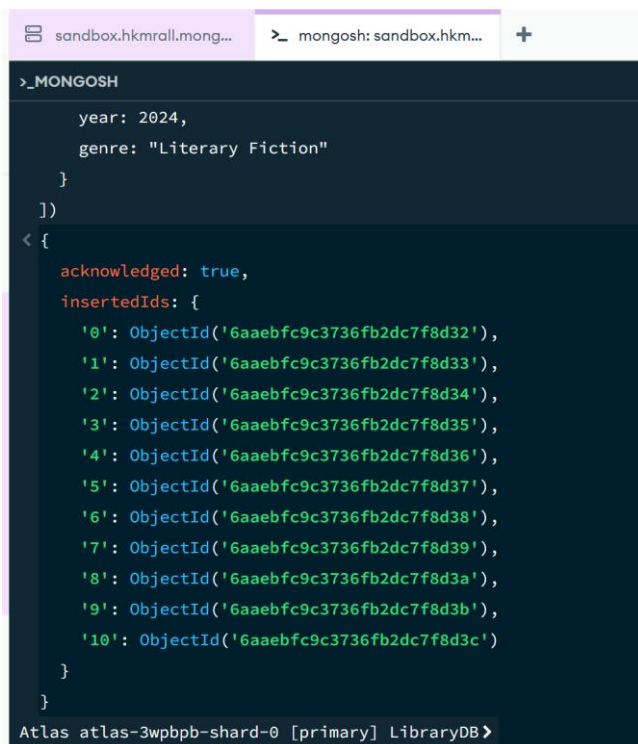
1. **Social media apps** — user-generated data (posts, likes, connections) is high-volume and structurally variable; document stores allow data that's frequently accessed together (e.g. a post, its comments, its like count) to be stored in the same place and retrieved as one unit, avoiding the multiple joins a relational schema and allowing for fast scaling
2. **Navigation systems** — location data needs fast, continuously updating "find nearest" or "within radius" queries (e.g., nearest petrol station, points along a route). MongoDB (NoSQL) supports this natively with GeoJSON storage and specialised geospatial indexes, whereas SQL databases need separate spatial extensions and are less suited to high-frequency location updates at scale due to the compute power required.

2. Working with MongoDB Collections and Documents (7 Marks)

a) Create a MongoDB database called LibraryDB.



b) Create a collection Books and insert at least three documents with fields like title, author, year, and genre.



c) Use the find command to fetch all books where year is greater than 2000.

```
> db.Books.find({year:{$gt:2000}}).pretty()
< {
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d35'),
  title: 'The Silent Patient',
  author: 'Alex Michaelides',
  year: 2019,
  genre: 'Thriller'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d36'),
  title: 'Tomorrow, and Tomorrow, and Tomorrow',
  author: 'Gabrielle Zevin',
  year: 2022,
  genre: 'Literary Fiction'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d37'),
  title: 'Fourth Wing',
  author: 'Rebecca Yarros',
  year: 2023,
  genre: 'Fantasy'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d38'),
  title: 'Lessons in Chemistry',
  author: 'Bonnie Garmus',
  year: 2022,
  genre: 'Historical Fiction'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d39'),
  title: 'Yellowface',
  author: 'R.F. Kuang',
  year: 2023,
  genre: 'Satire'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d3a'),
  title: 'The Housemaid',
  author: 'Freida McFadden',
  year: 2022,
  genre: 'Thriller'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d3b'),
  title: 'Iron Flame',
  author: 'Rebecca Yarros',
  year: 2023,
  genre: 'Fantasy'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d3c'),
  title: 'James',
  author: 'Percival Everett',
  year: 2024,
  genre: 'Literary Fiction'
}
```

3. Updating and Deleting Documents in MongoDB (5 Marks)

a) Use the LibraryDB database and update the genre of the book titled "Book1" to "Classic Fiction".

```
> db.Books.find({"title":"Book1"},{title:1, genre:1, _id:0})
< {
  title: 'Book1',
  genre: 'Science Fiction'
}
> db.Books.findOneAndUpdate({ title: "Book1" },{ $set: { genre: "Classic Fiction" } },
  { projection: { title: 1, genre: 1, _id: 0 } })
< {
  title: 'Book1',
  genre: 'Science Fiction'
}
```

b) Delete any book in the collection where the year is less than 2000.

```
> db.Books.find({ year: { $lt: 2020 } }).pretty()
```

```
< {
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d32'),
  title: 'Book1',
  author: 'Frank Herbert',
  year: 1965,
  genre: 'Classic Fiction'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d33'),
  title: 'The Hobbit',
  author: 'J.R.R. Tolkien',
  year: 1937,
  genre: 'Fantasy'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d34'),
  title: '1984',
  author: 'George Orwell',
  year: 1949,
  genre: 'Dystopian'
}
{
  _id: ObjectId('6aaebfc9c3736fb2dc7f8d35'),
  title: 'The Silent Patient',
  author: 'Alex Michaelides',
  year: 2019,
  genre: 'Thriller'
}
```

```
> db.Books.find({ year: { $lt: 2020 } }).count()
```

```
< 3
```

```
> db.Books.deleteMany({ year: { $lt: 2000 } })
```

```
< {
  acknowledged: true,
  deletedCount: 3
}
```

c) Use find() to verify the updates and deletions.

```
> db.Books.find({ title: "Book1" }, { title: 1, genre: 1, _id: 0 })  
< {  
  title: 'Book1',  
  genre: 'Classic Fiction'  
}
```

```
> db.Books.find({ year: { $lt: 2000 } }).pretty()  
<  
> db.Books.find({ year: { $lt: 2000 } }).count()  
< 0
```